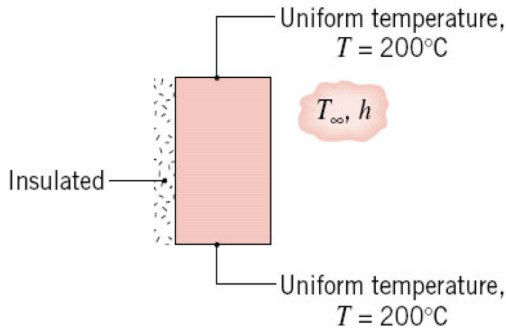


CFD HW #5 2024109 권혁술

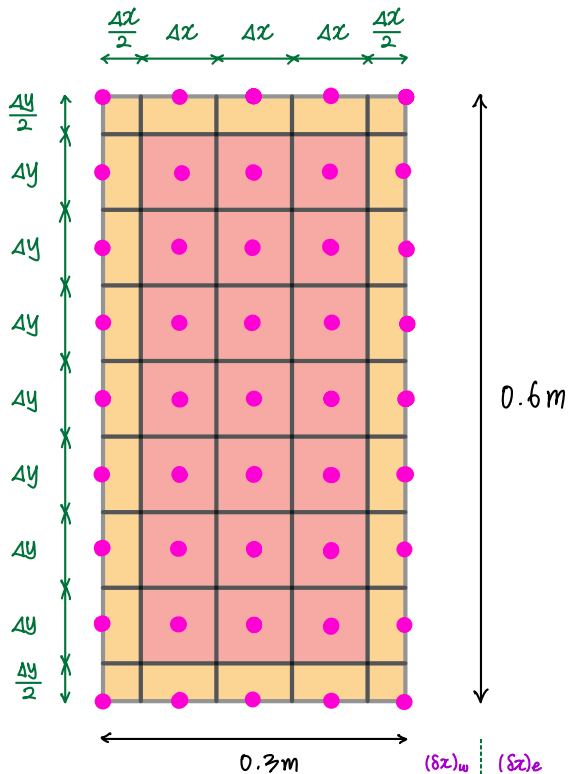
1. A long bar of rectangular cross section, $0.3 \text{ m} \times 0.6 \text{ m}$ on a side and having a thermal conductivity of 2.0 W/m-K , is subjected to the boundary conditions shown.



→ 2D steadystate conduction

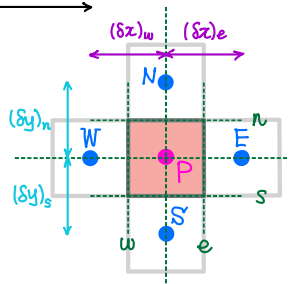
Two of the sides are maintained at a uniform temperature of 200°C . One of the sides is adiabatic, and the remaining side is subjected to a convection process with $h = 20 \text{ W/m}^2\text{-K}$ and $T_\infty = 25^\circ\text{C}$ and. Using an appropriate numerical technique (SOR), determine the temperature distribution in the bar and the heat transfer rate between the bar and the fluid per unit length of the bar.

i) Step 1. Discretization of the domain



$$\Delta x = \frac{0.3 \text{ m}}{4} = 0.075 \text{ m}$$

$$\Delta y = \frac{0.6 \text{ m}}{8} = 0.075 \text{ m}$$



ii) Step 2. Integration of the governing differential equation

$$\frac{\partial}{\partial t}(\rho \phi) + \nabla \cdot (\rho \vec{u} \phi) = \nabla \cdot (\Gamma \nabla \phi) + S_\phi$$

$$\rightarrow \nabla \cdot (k \nabla T) = 0$$

$$\int_s^e \int_w^e \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) dx dy + \int_s^e \int_w^e \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) dx dy = 0$$

$$\left[\left(k \frac{\partial T}{\partial x} \right)_e - \left(k \frac{\partial T}{\partial x} \right)_w \right] \Delta y + \left[\left(k \frac{\partial T}{\partial y} \right)_n - \left(k \frac{\partial T}{\partial y} \right)_s \right] \Delta x = 0$$

iii) Selection of piece wise profile

I selected piecewise linear profile for considering the gradient of T .

$$\rightarrow \left[k_e \frac{T_E - T_P}{(\delta x)_e} - k_w \frac{T_P - T_W}{(\delta x)_w} \right] \Delta y + \left[k_n \frac{T_N - T_P}{(\delta y)_n} - k_s \frac{T_P - T_S}{(\delta y)_s} \right] \Delta x = 0$$

$$\left\{ \frac{k_e \Delta y}{(\delta x)_e} + \frac{k_w \Delta y}{(\delta x)_w} + \frac{k_n \Delta z}{(\delta x)_n} + \frac{k_s \Delta z}{(\delta x)_s} \right\} T_p = \frac{k_e \Delta y}{(\delta x)_e} T_E + \frac{k_w \Delta y}{(\delta x)_w} T_W + \frac{k_n \Delta z}{(\delta x)_n} T_N + \frac{k_s \Delta z}{(\delta x)_s} T_S$$

$\downarrow$
 $a_p (= a_E + a_W + a_N + a_S)$

$\downarrow$
 $a_E (>0)$

$\downarrow$
 $a_W (>0)$

$\downarrow$
 $a_N (>0)$

$\downarrow$
 $a_S (>0)$

$$\longrightarrow a_p T_p = a_E T_E + a_W T_W + a_N T_N + a_S T_S \quad : \text{ discretized equation}$$

* The 4 basic rules .

1) Rule #1 . Consistency at control volume faces : *Satisfied*

→ The fluxes across the entire control volume surfaces are consistent since I used a piecewise linear profile to represent the spatial variation of temperature and the thermal conductivity is constant throughout the domain.

2) Rule #2. Positive coefficients : *Satisfied*

→ All coefficients are positive .

$$a_E > 0 , a_W > 0 , a_N > 0 , a_S > 0 \longrightarrow a_p = a_E + a_W + a_N + a_S > 0$$

3) Rule #3. Negative - slope linearization of the source term : *Satisfied*

→ In our problem, no source is present, so the rule is always satisfied.

4) Rule #4. Sum of the neighbor coefficients : *Satisfied*

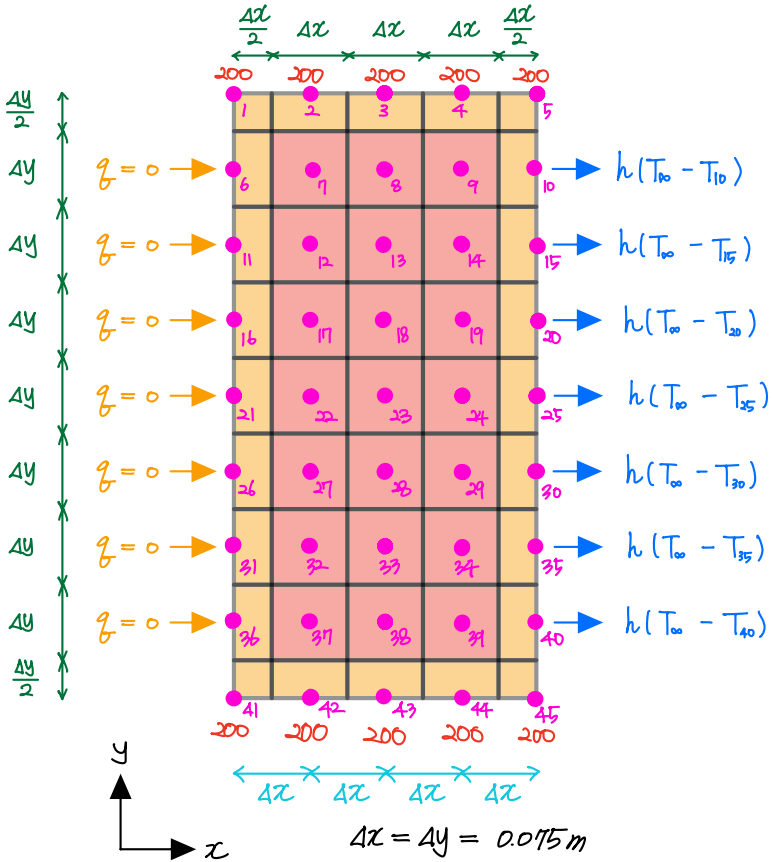
→ In our problem, the differential equation has only derivative term. So, $a_p = \sum a_{nb}$ continues to remain satisfied after a constant is added to the dependent variable.

* Scarborough criterion

I already showed that $a_p = \sum a_{nb}$ in the entire domain . (We don't have any source here.)

Thus, $\frac{\sum |a_{nb}|}{|a_p|} \leq 1$ is satisfied for all equations .

* Boundary conditions



4) Internal nodes update

* SOR (Successive Over relaxation) method

Our discretization equation is $A_p T_p = \sum A_{nb} T_{nb}$.

We can also write it as follows.

$$T_p = T_p^* + \left(\frac{\sum A_{nb} T_{nb}}{A_p} - T_p^* \right)$$

Introducing a relaxation parameter, α

$$T_p = T_p^* + \alpha \left(\frac{\sum A_{mb} T_{mb}}{a_p} - T_p^* \right) = \frac{\alpha}{a_p} (\sum A_{mb} T_{mb}) + (1 - \alpha) T_p^*$$

For 2D, the SOR method is given by

$$T_P^t = \frac{\alpha}{a_p} (a_{E T_E}^{t-1} + a_{W T_W}^t + a_{N T_N}^{t-1} + a_{S T_S}^t) + (1-\alpha) T_P^{t-1}$$

1) Top and Bottom

$$\begin{aligned} T_1 \sim T_5 &= T_{\text{fixed}} = 200 \\ T_{41} \sim T_{45} &= T_{\text{fixed}} = 200 \end{aligned}$$

2) Left boundary

$$k \left(\frac{T_7 - T_6}{\Delta x} \right) = 0 \rightarrow T_6 = T_7$$

$$K \left(\frac{T_{12} - T_{11}}{4x} \right) = 0 \rightarrow T_{11} = T_{12}$$

$$k \left(\frac{T_{37} - T_{36}}{4x} \right) = 0 \rightarrow T_{36} = T_{37}$$

3) Right boundary

$$h(T_{\infty} - T_{10}) - k \left(\frac{T_{10} - T_q}{\Delta x} \right) = 0$$

$$\rightarrow T_{10} = \frac{\left(\frac{k}{\Delta x}\right) T_9 + h T_{\infty}}{(h + \frac{k}{\Delta x})}$$

$$h(T_{\infty} - T_{20}) - k \left(\frac{T_{20} - T_{19}}{\Delta x} \right) = 0$$

$$\rightarrow T_{20} = \frac{(k/\Delta x) T_{19} + h T_{\infty}}{(h + k/\Delta x)}$$

$$h(T_{\infty} - T_{40}) - k \left(\frac{T_{40} - T_{39}}{\Delta x} \right) = 0$$

$$\rightarrow T_{40} = \frac{(K/\Delta x) T_{39} + h T_{\infty}}{(h + K/\Delta x)}$$

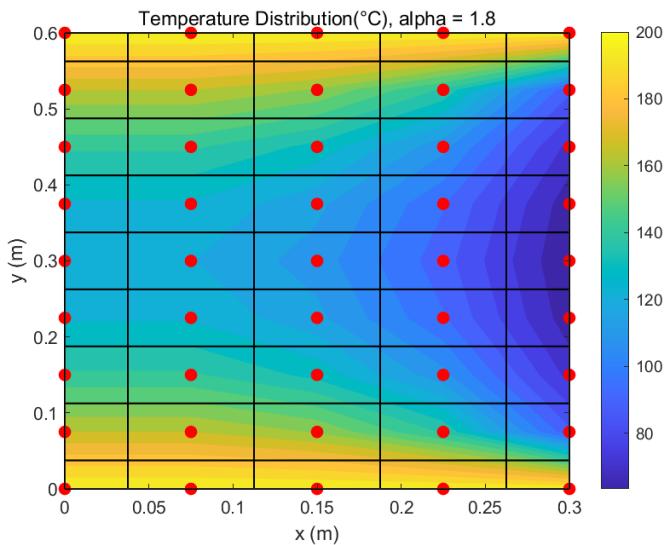
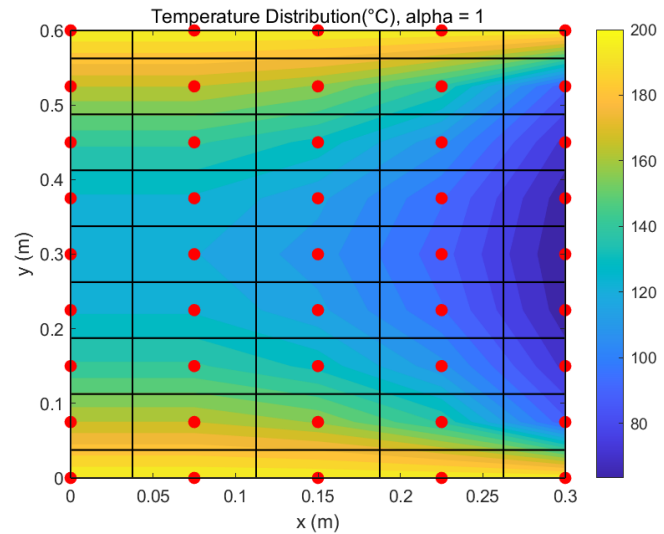
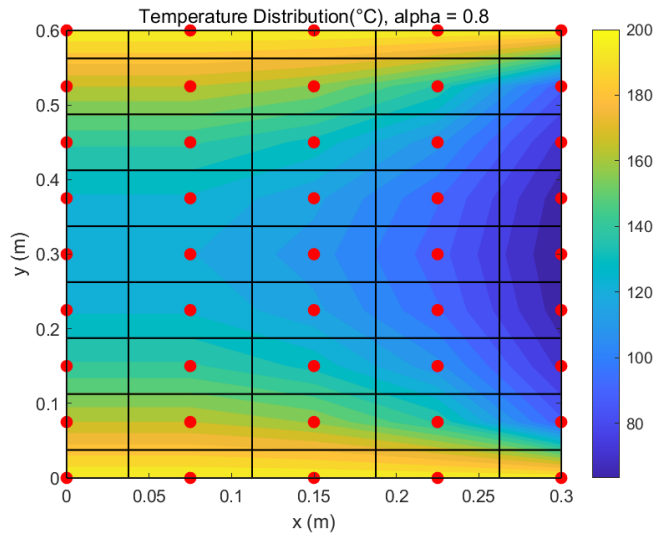
< MATLAB >

1	%% 2D Steady Heat Conduction with SOR	
2	% 20244109 Hyeoksul Kwon	
3		
4	clear; clc; close all;	
5	%% Domain and grid parameters	
6	Lx = 0.3; % total length in x-direction (m)	
7	Ly = 0.6; % total length in y-direction (m)	
8	dx = 0.075; % (del x)_e,w	
9	dy = 0.075; % (del y)_n,s	
10		
11	Nx = Lx/(dx) + 0.5*2 % 5 nodes in x	
12	Ny = Ly/(dy) + 0.5*2 % 9 nodes in y	
13		
14	%% Physical parameters	
15	k = 2.0; % thermal conductivity (W/(m.K))	
16	h = 20; % convection heat transfer coeff (W/(m^2.K))	
17	T_inf = 25; % ambient temperature (°C)	
18	T_fixed = 200; % Dirichlet BC at top/bottom	
19		
20	%% SOR parameters	
21	alpha_values = [1, 1.8, 0.8];	
22	tol = 1e-6;	
23	max_iter = 1000;	
24		
25	%% To store iteration counts and heat transfer results	
26	iter_counts = zeros(length(alpha_values), 1);	
27	heat_rate = zeros(length(alpha_values), 1); % total heat transfer rate per unit length at right boundary	
28		
29	for a = 1:length(alpha_values)	
30	alpha = alpha_values(a);	
31		
32	% Initialize T as 20°C everywhere	
33	T = ones(Ny, Nx) * 20;	
34		
35	% Apply Dirichlet BC at top (j=Ny) and bottom (j=1)	
36	T(1, :) = T_fixed; % bottom	
37	T(Ny, :) = T_fixed; % top	
38		
39	iter = 0;	
40	err = inf;	
41		
42	% Precompute A = k/delta_x for convection BC	
43	A = k/dx;	
44		
45	%% SOR iterative loop	
46	while err > tol && iter < max_iter	
47	iter = iter + 1;	
48	T_old = T;	
49		
50	% Update internal nodes(SOR)	
51	% column j = 2..Ny-1, row i = 2..Nx-1	
52	for j = 2:(Ny-1)	
53	for i = 2:(Nx-1)	
54	% East (i+1), North (j+1) => T_old	
55	% West (i-1), South (j-1) => T (already updated in current iteration)	
56		
57	T_new = (T_old(j, i+1) + T_old(j+1, i) + T(j, i-1) + T(j-1, i)) / 4;	
58	T(j,i) = (1 - alpha)*T_old(j,i) + alpha*T_new;	
59	end	
60	end	
61		
62	% Left boundary (insulated)	
63	% T(j,1) = T(j,2) for j=2..Ny-1	
64	for j = 2:(Ny-1)	
65	T(j,1) = T(j,2);	
66	end	
67		
68	% Right boundary (convection)	
69	% T(j,Nx) = [A*T(j,Nx-1) + h*T_inf] / (A + h)	
70	for j = 2:(Ny-1)	
71	T(j,Nx) = (A*T(j,Nx-1) + h*T_inf) / (A + h);	
72	end	
73		
74	% Re-apply Dirichlet BC at top/bottom	
75	T(1,:) = T_fixed;	
76	T(Ny,:) = T_fixed;	
77		
78	% Calculate max error	
79	err = max(max(abs(T - T_old)));	
80		
81	end	
82		
83	iter_counts(a) = iter;	
84	fprintf('alpha = %g converged in %d iterations, final err = %e\n', alpha, iter, err);	
85		
86	%% Compute heat transfer rate per unit length from right boundary	
87	% For each interior node j=2..(Ny-1) at x=Lx (i=Nx):	
88	% q_j = h * (T(j,Nx) - T_inf) * dy	
89	% Summation of q_j over j=2..Ny-1 => total heat rate per unit length(W/m).	
90	q = zeros(1,Ny);	
91	for j = 2:(Ny-1)	
92	q(j) = h * (T(j,Nx) - T_inf)*dy;	
93	end	
94		
95	q_dot = sum(q)	
96		
97	% Save this heat rate result	
98	heat_rate(a) = q_dot;	
99		
100	fprintf('alpha = %g: iterations = %d, Q_dot = %f W/m\n', alpha_values(a), iter_counts(a), heat_rate(a));	
101		

$$\dot{q} = \sum_{j=2}^{N_y-2} h(T_{j,N_x} - T_{\infty}) \times 4y$$

* Result and Discussion

1) Using 14 cells



```
Nx =
    5

Ny =
    9

alpha = 1 converged in 143 iterations, final err = 9.110119e-07

q_dot =
    526.1124

alpha = 1: iterations = 143, Q_dot = 526.112419 W/m
alpha = 1.8 converged in 41 iterations, final err = 6.859301e-07

q_dot =
    526.1124

alpha = 1.8: iterations = 41, Q_dot = 526.112437 W/m
alpha = 0.8 converged in 202 iterations, final err = 9.573277e-07

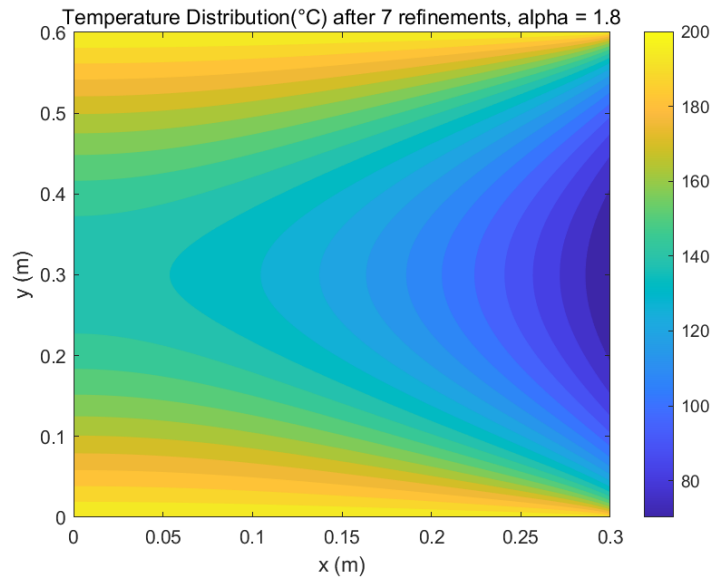
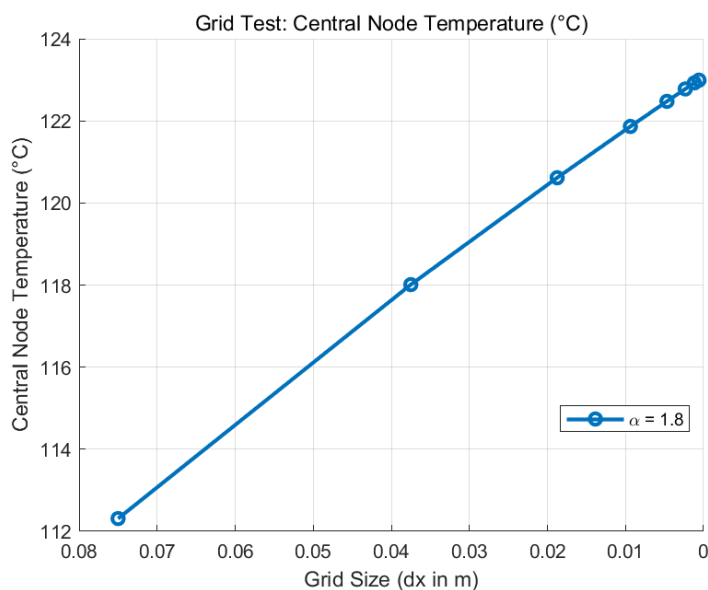
q_dot =
    526.1124

alpha = 0.8: iterations = 202, Q_dot = 526.112408 W/m
Iteration counts and heat transfer:
alpha = 1: iterations = 143, Q_dot = 526.112419 W/m
alpha = 1.8: iterations = 41, Q_dot = 526.112437 W/m
alpha = 0.8: iterations = 202, Q_dot = 526.112408 W/m
>>
```

Firstly, the numerical results obtained using 14 cells are as shown above. When using SOR, the solution was computed using an alpha value of 0.8 for under relaxation, 1 for the Gauss-Seidel method, and 1.8 for over relaxation. As a result, when alpha = 1.8 was used, the fewest iterations were required (41 iterations). In contrast, alpha = 1 resulted in 143 iterations, and alpha = 0.8 resulted in 202 iterations.

It was observed that a larger alpha value led to fewer iterations, confirming that using alpha = 1.8 was the optimal choice. Additionally, for all three cases, the heat transfer rate per unit length between the bar and the fluid was calculated as 526.11 W/m. Since these results were unreliable, i then conducted grid independence test.

2) Grid independence test and Final result



```
Alpha = 1.8, Refinement Level 0: dx = 0.075, iterations = 41, final err = 6.859301e-07
Alpha = 1.8, Refinement Level 1: dx = 0.0375, iterations = 75, final err = 8.037694e-07
Alpha = 1.8, Refinement Level 2: dx = 0.01875, iterations = 277, final err = 9.495918e-07
Alpha = 1.8, Refinement Level 3: dx = 0.009375, iterations = 959, final err = 9.944234e-07
Alpha = 1.8, Refinement Level 4: dx = 0.0046875, iterations = 3341, final err = 9.981370e-07
Alpha = 1.8, Refinement Level 5: dx = 0.00234375, iterations = 11705, final err = 9.997566e-07
Alpha = 1.8, Refinement Level 6: dx = 0.00117187, iterations = 40928, final err = 9.998095e-07
Alpha = 1.8, Refinement Level 7: dx = 0.000585937, iterations = 141641, final err = 9.999469e-07
```

Grid Test Results:

Alpha = 1.8:

```
Level 0: dx = 0.075, Central Temp = 112.311098 °C
Level 1: dx = 0.0375, Central Temp = 118.010518 °C
Level 2: dx = 0.01875, Central Temp = 120.615205 °C
Level 3: dx = 0.009375, Central Temp = 121.863829 °C
Level 4: dx = 0.0046875, Central Temp = 122.475496 °C
Level 5: dx = 0.00234375, Central Temp = 122.777802 °C
Level 6: dx = 0.00117187, Central Temp = 122.926193 °C
Level 7: dx = 0.000585937, Central Temp = 122.992100 °C
```

Iteration counts and heat transfer:

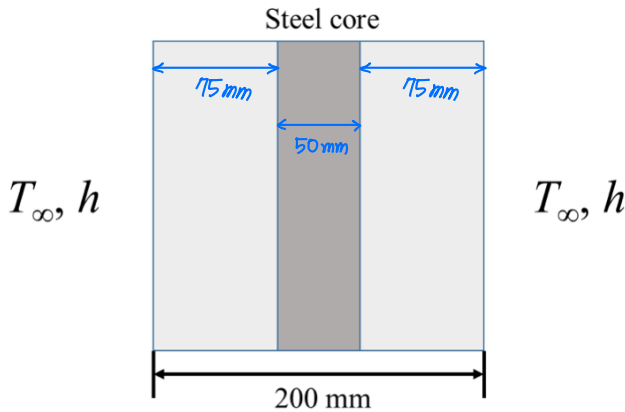
```
alpha = 1.8: iterations = 141641, Q_dot = 835.930866 W/m
```

Since a relatively small number of grid cells were used previously, it was expected that the solution error would be significant. Therefore, a grid test was conducted. The grid size was set starting at 0.075 m and was halved step by step according to the refinement level. A grid independence test was performed by specifying the temperature at the center of the domain. I used 1.8 for the value of alpha. The results showed that the solution nearly converged when the grid size was reduced below 0.001m, and at a grid size of 0.000585937m, an error of approximately 0.0659 was observed compared to the previous grid size, indicating that the temperature values had converged with respect to the grid size.

Following the grid independence test, the numerical solution was recalculated using a grid size of 0.000585937, which consisted of 525,825 cells. An alpha value of 1.8 was used for the simulation.

As a result, 141,641 iterations were performed, and the heat transfer rate per unit length between the bar and the fluid was computed as 835.93 W/m *Answer*

2. Steel-reinforced concrete pillars are used in the construction of large buildings. Structural failure can occur at high temperatures due to a fire because of softening of the metal core. Consider a 200-mm-thick composite pillar consisting of a central steel core (50 mm thick) sandwiched between two 75-mm-thick concrete walls. The pillar is at a uniform initial temperature of $T_i = 27^\circ\text{C}$ and is suddenly exposed to combustion products at $T_\infty = 900^\circ\text{C}$, $h = 40 \text{ W/m}^2\text{-K}$ on both exposed surfaces. The surroundings temperature is also 900°C .

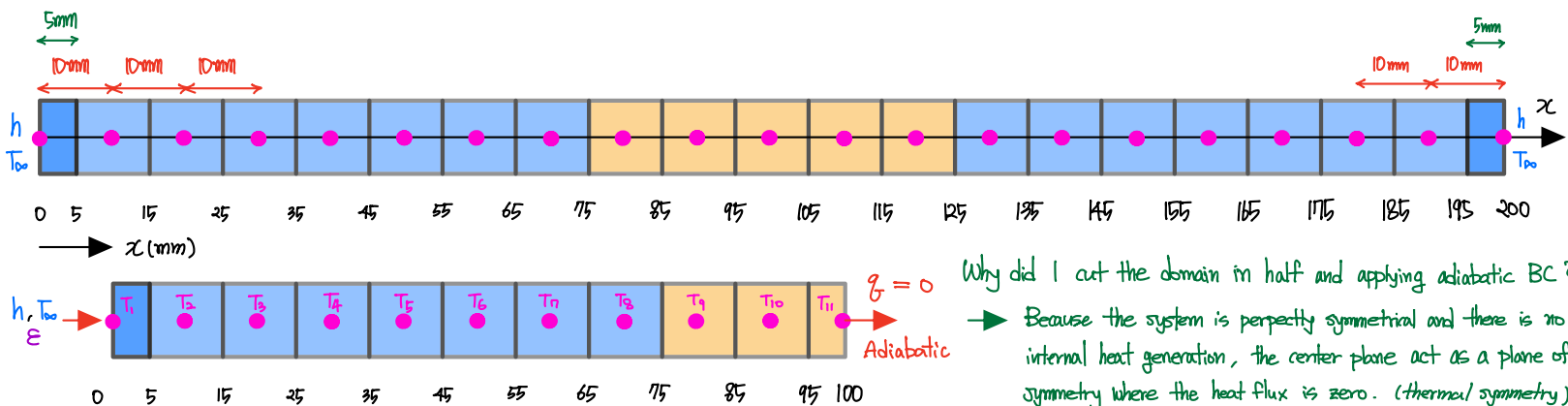


→ 1D unsteady conduction

$$\rho c \frac{dT}{dt} = \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right)$$

Using an implicit finite difference method with proper selection of Δx and Δt , determine the temperature of the exposed concrete surface and the center of the steel plate at $t = 10,000 \text{ s}$. Steel properties are: $k_s = 55 \text{ W/m-K}$, $\rho_s = 7850 \text{ kg/m}^3$, and $c_s = 450 \text{ J/kg-K}$. Concrete properties are: $k_c = 1.4 \text{ W/m-K}$, $\rho_c = 2300 \text{ kg/m}^3$, $c_c = 880 \text{ J/kg-K}$, and $\varepsilon = 0.90$. Plot the maximum and minimum concrete temperatures along with the maximum and minimum steel temperatures over the the duration $0 \leq t \leq 10,000 \text{ s}$.

i) Step 1. Discretization of the domain



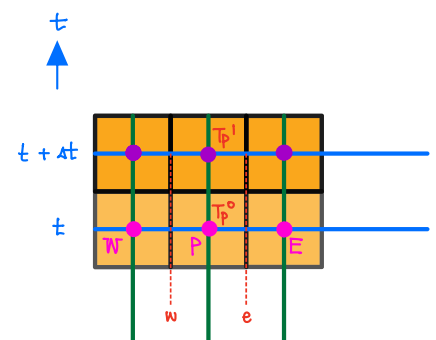
ii) Step 2. Integrating the governing equation over the CV and over the time interval.

$$\rho c \int_w^e \int_t^{t+\Delta t} \frac{\partial T}{\partial t} dt dz = \int_w^e \int_t^{t+\Delta t} \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) dt dz$$

Assuming that T_p prevails throughout the CV

$$\text{LHS: } \rho c \int_w^e (T^{t+\Delta t} - T^t) dz = \rho c (T_p^{t+\Delta t} - T_p^t) \Delta x$$

($T_p^{t+\Delta t}$: the new unknown values at $t + \Delta t$, T_p^t : the old value at t)



iii) Step 3. Selecting a proper piecewise profile

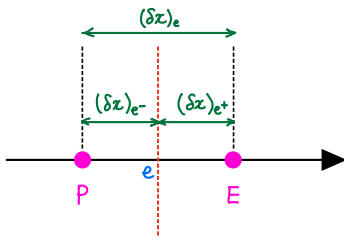
→ I selected a piecewise linear profile to consider a gradient of the temperature.

$$\text{RHS} : \int_w^{t+\Delta t} \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) dt dx = \int_t^{t+\Delta t} \left[k_e \frac{T_E - T_P}{(\delta x)_e} - k_w \frac{T_P - T_W}{(\delta x)_w} \right] dt$$

At the interface, the thermal conductivity is determined using the harmonic mean to account for both the consistency at the control volume interface and the continuous variation of conductivity between different variables.

* The interface conductivity

by electric circuit analogy,



$$g_e = \frac{T_P - T_E}{(\delta x)_e^- / k_P + (\delta x)_e^+ / k_E} \rightarrow \frac{(\delta x)_e}{k_e} = \frac{(\delta x)_e^-}{k_P} + \frac{(\delta x)_e^+}{k_E}$$

$$k_e = \left(\frac{1 - f_e}{k_P} + \frac{f_e}{k_E} \right)^{-1}$$

$$(f_e \equiv \frac{(\delta x)_e^+}{(\delta x)_e}, \quad 1 - f_e \equiv \frac{(\delta x)_e^-}{(\delta x)_e})$$

In the current discretized domain, the value of f on the surface "e" is as follows.

$$(\delta x)_e = 10 \text{ mm}, \quad (\delta x)_e^+ = 5 \text{ mm}$$

$$f_e = \frac{(\delta x)_e^+}{(\delta x)_e} = \frac{1}{2}$$

$$\rightarrow k_e = \left(\frac{1/2}{k_P} + \frac{1/2}{k_E} \right)^{-1} = \frac{2k_P k_E}{k_E + k_P}$$

According to the problem, we have to use implicit method.

$$\rho C (T_P^1 - T_P^0) \Delta x = \int_t^{t+\Delta t} \left[k_e \frac{T_E - T_P}{(\delta x)_e} - k_w \frac{T_P - T_W}{(\delta x)_w} \right] dt = \left[k_e \frac{T_E^1 - T_P^1}{(\delta x)_e} - k_w \frac{T_P^1 - T_W^1}{(\delta x)_w} \right] \Delta t$$

Dropping the superscript 1

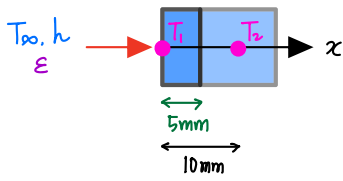
$$\left(\underbrace{\frac{k_e}{(\delta x)_e} + \frac{k_w}{(\delta x)_w} + \frac{\rho C \cdot \Delta x}{\Delta t}}_{a_P} \right) T_P = \underbrace{\frac{k_e}{(\delta x)_e}}_{a_E} T_E + \underbrace{\frac{k_w}{(\delta x)_w}}_{a_W} T_W + \underbrace{\frac{\rho C \cdot \Delta x}{\Delta t} T_P^0}_b$$

$$\rightarrow a_P T_P = a_E T_E + a_W T_W + b \quad (a_P = a_E + a_W + a_P^0, \quad b = a_P^0 T_P^0)$$

$$a_E > 0, \quad a_W > 0, \quad a_P^0 > 0 \rightarrow a_P > 0$$

$$\epsilon = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

* left boundary

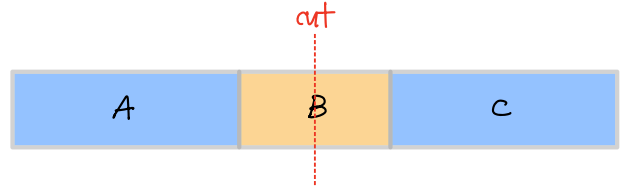


$$\rho_c c_c (T_1 - T_1^0) \cdot (0.0005) = \left[\frac{k_e (T_2 - T_1)}{(0.001)} - \{ h (T_1 - T_\infty) + \epsilon \sigma (T_1^4 - T_\infty^4) \} \right] \cdot \Delta t$$

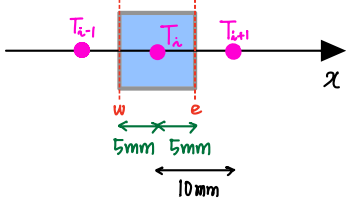
$$k_e = \frac{2 \cdot k_c k_c}{k_c + k_c} = k_c \text{ (harmonic mean)}$$

$$\frac{k_e}{(0.001)} T_2 = \left\{ \frac{\rho_c c_c}{\Delta t} (0.0005) + \frac{k_e}{(0.001)} + h \right\} T_1 - h T_\infty - \frac{\rho_c c_c T_1^0 \cdot (0.0005)}{\Delta t} + \epsilon \sigma (T_1^4 - T_\infty^4)$$

* Internal nodes



i) Region A ($T_i = T_2 \sim T_7$)

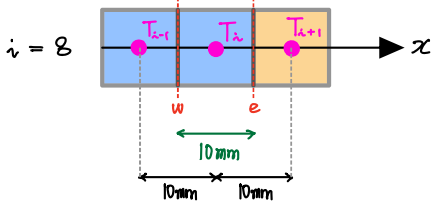


$$\rho_c c_c (T_i - T_i^0) \cdot (0.001) = \left[\frac{k_e (T_{i+1} - T_i)}{(0.001)} - \frac{k_w (T_i - T_{i-1})}{(0.001)} \right] \cdot \Delta t$$

$$k_e = k_w = \frac{2 \cdot k_c k_c}{k_c + k_c} = k_c \text{ (harmonic mean)}$$

$$\frac{k_e}{(0.001)} T_{i+1} = \left\{ \frac{k_e}{(0.001)} + \frac{k_w}{(0.001)} + \frac{\rho_c c_c \cdot (0.001)}{\Delta t} \right\} T_i - \frac{k_w}{0.001} T_{i-1} - \frac{\rho_c c_c \cdot (0.001)}{\Delta t} \cdot T_i^0$$

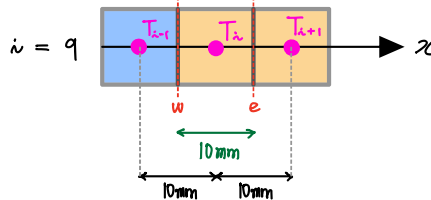
ii) Region A ~ B ($T_i = T_8, T_9$)



$$\rho_c c_c (T_i - T_i^0) \cdot (0.001) = \left[\frac{k_e (T_{i+1} - T_i)}{(0.001)} - \frac{k_w (T_i - T_{i-1})}{(0.001)} \right] \cdot \Delta t$$

$$k_w = \frac{2 \cdot k_c k_c}{k_c + k_c} = k_c, \quad k_e = \frac{2 \cdot k_c k_s}{k_c + k_s} \text{ (harmonic mean)}$$

$$\frac{k_e}{(0.001)} T_{i+1} = \left\{ \frac{k_e}{(0.001)} + \frac{k_w}{(0.001)} + \frac{\rho_c c_c \cdot (0.001)}{\Delta t} \right\} T_i - \frac{k_w}{0.001} T_{i-1} - \frac{\rho_c c_c \cdot (0.001)}{\Delta t} \cdot T_i^0$$

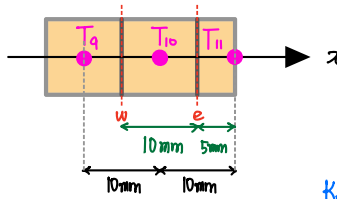


$$\rho_s c_s (T_i - T_i^0) \cdot (0.001) = \left[\frac{k_e (T_{i+1} - T_i)}{(0.001)} - \frac{k_w (T_i - T_{i-1})}{(0.001)} \right] \cdot \Delta t$$

$$k_w = \frac{2 \cdot k_c k_s}{k_c + k_s}, \quad k_e = \frac{2 \cdot k_s k_s}{k_s + k_s} = k_s \text{ (harmonic mean)}$$

$$\frac{k_e}{(0.001)} T_{i+1} = \left\{ \frac{k_e}{(0.001)} + \frac{k_w}{(0.001)} + \frac{\rho_s c_s \cdot (0.001)}{\Delta t} \right\} T_i - \frac{k_w}{0.001} T_{i-1} - \frac{\rho_s c_s \cdot (0.001)}{\Delta t} \cdot T_i^0$$

iii) Region B

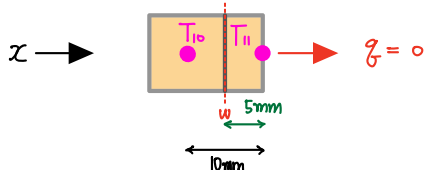


$$\rho_s c_s (T_{10} - T_{10}^0) \cdot (0.001) = \left[\frac{k_e (T_{11} - T_{10})}{(0.001)} - \frac{k_w (T_{10} - T_q)}{(0.001)} \right] \cdot \Delta t$$

$$k_e = k_w = \frac{2 \cdot k_s k_s}{k_s + k_s} = k_s \text{ (harmonic mean)}$$

$$\frac{k_e}{(0.001)} T_{11} = \left\{ \frac{k_e}{(0.001)} + \frac{k_w}{(0.001)} + \frac{\rho_c c_c \cdot (0.001)}{\Delta t} \right\} T_{10} - \frac{k_w}{0.001} T_q - \frac{\rho_c c_c \cdot (0.001)}{\Delta t} \cdot T_{10}^0$$

* Right boundary



$$\rho_s c_s (T_{11} - T_{11}^0) \cdot (0.0005) = \left[0 - \frac{k_w (T_{11} - T_{10})}{(0.001)} \right] \cdot \Delta t \quad k_w = \frac{2 \cdot k_s k_s}{k_s + k_s} = k_s \text{ (harmonic mean)}$$

$$\left(\frac{\rho_s c_s (0.0005)}{\Delta t} + \frac{k_w}{(0.001)} \right) T_{11} = \frac{k_w}{(0.001)} T_{10} + \frac{\rho_s c_s (0.0005)}{\Delta t} T_{11}^0$$

- The four basic rules

1) Rule #1 : Consistency at control volume surfaces

→ It is satisfied since I used harmonic mean conductivity formulation over the entire interfaces.

2) Rule #2 : Positive coefficients ($a_p, a_E, a_W > 0$)

→ It is satisfied since a_E, a_W , and a_p° are always positive.

$$a_p = a_E + a_W + a_p^\circ > 0$$

3) Rule #3 : Negative - slope linearization

→ It is satisfied since we don't have any source ($\bar{S} = 0$)

4) Sum of the neighbor coefficients ($a_p = \sum a_{nb}$)

→ It is satisfied since the differential equation we are dealing with is as follows.

$$\rho c \frac{dT}{dt} = \frac{d}{dx} \left(k \frac{dT}{dx} \right)$$

We only have derivative term and don't have any source term.

So the differential equation continues to remain satisfied after a constant is added to the dependant variable.

Thus, $a_p = \sum a_{nb}$ is always satisfied.

- Scarborough criterion

$$a_E = \frac{k_e}{(\delta x)_e}, \quad a_W = \frac{k_w}{(\delta x)_w}, \quad a_p^\circ = \frac{\rho c A \Delta x}{\Delta t}, \quad a_p = a_E + a_W + a_p^\circ \rightarrow a_p T_p = a_E T_E + a_W T_W + a_p^\circ T_p^\circ$$

$a_p = \sum a_{nb}$ (spatial + temporal neighbor)

$$\frac{\sum |a_{nb}|}{|a_p|} \leq 1 \text{ is always satisfied. So, it is sufficient for convergence.}$$

* Step 4. Solve the discretized equation.

$$a_p T_p = a_E T_E + a_W T_W + b \rightarrow \text{TDMA Algorithm}$$

<MATLAB>

```
1 %% 1D Transient Heat Conduction
2 % 20244109 Hyeoksul Kwon
3
4 clear; clc; close all;
5
6 %% Parameters and Grid Generation
7
8 L_total_mm = 100; % [mm] half-domain from 0 to 100 mm
9
10 % ** USER INPUT: Set the total number of nodes **
11 N_nodes = 1281;
12
13 % Generate node positions (in mm) uniformly between 0 and L_total_mm.
14 x_node_mm = linspace(0, L_total_mm, N_nodes)'; % Column vector, in mm
15
16 % Compute the uniform spacing between nodes (in mm)
17 dx_local_mm = x_node_mm(2) - x_node_mm(1);
18
19 % Create cell widths (w_mm) using the same rule:
20 % First and last cells are half cells (dx_local_mm/2), the rest are full cells.
21 w_mm = zeros(N_nodes,1);
22 w_mm(1) = dx_local_mm/2;
23 w_mm(2:end-1) = dx_local_mm;
24 w_mm(end) = dx_local_mm/2;
25
26 % Convert node positions and cell widths to meters
27 x_node = x_node_mm * 1e-3; % [m]
28 w = w_mm * 1e-3; % [m]
29
30 %% Material Properties
31
32 % Concrete : k = 1.4 W/(m·K), rho = 2300 kg/m^3, c = 880 J/(kg·K)
33 % Steel : k = 55 W/(m·K), rho = 7850 kg/m^3, c = 450 J/(kg·K)
34 k_conc = 1.4; rho_conc = 2300; c_conc = 880;
35 k_steel = 55.0; rho_steel = 7850; c_steel = 450;
36
37 % Allocate material properties based on x position (in mm)
38 % Outer region: x < 75 mm --> Concrete
39 % Inner region: x >= 75 mm --> Steel
40 k_arr = zeros(N_nodes, 1);
41 rho_arr = zeros(N_nodes, 1);
42 c_arr = zeros(N_nodes, 1);
43 for i = 1:N_nodes
44     if x_node_mm(i) < 75
45         k_arr(i) = k_conc;
46         rho_arr(i) = rho_conc;
47         c_arr(i) = c_conc;
48     else
49         k_arr(i) = k_steel;
50         rho_arr(i) = rho_steel;
51         c_arr(i) = c_steel;
52     end
53 end
54
55 %% Boundary Conditions and Other Parameters
56
57 % Node 1 (x = 0 mm): Convection + Radiation
58 % Last Node (x = L_total_mm): Adiabatic (no external flux)
59 T_inf_C = 900; % Ambient temperature in Celsius
60 T_inf = T_inf_C + 273.15; % [K]
61 h = 40.0; % Convection coefficient [W/(m^2·K)]
62 epsilon = 0.9; % Surface emissivity
63 sigma = 5.67e-8; % Stefan-Boltzmann constant [W/(m^2·K^4)]
64
65 % Initial Temperature
66 T_init_C = 27; % Initial temperature in Celsius
67 T_init = T_init_C + 273.15; % [K]
68
69 % Time stepping parameters
70 t_end = 10000; % Final time [s]
71 dt = 1.0; % Time step [s]
72 t = 0; % Initial time
73 max_steps = ceil(t_end/dt);
74
75 % Preallocate arrays for time history
76 time_history = zeros(max_steps+1, 1);
77 T_history = zeros(max_steps+1, N_nodes);
78
79 % Identify regions based on node positions (in mm):
80 % Concrete region: nodes where x_node_mm < 75 mm
81 % Steel region: nodes where x_node_mm >= 75 mm
82 concrete_idx = find(x_node_mm < 75);
83 steel_idx = find(x_node_mm >= 75);
84 min_conc_T_hist = zeros(max_steps+1, 1);
85 max_conc_T_hist = zeros(max_steps+1, 1);
86 min_steel_T_hist = zeros(max_steps+1, 1);
87 max_steel_T_hist = zeros(max_steps+1, 1);
88
```

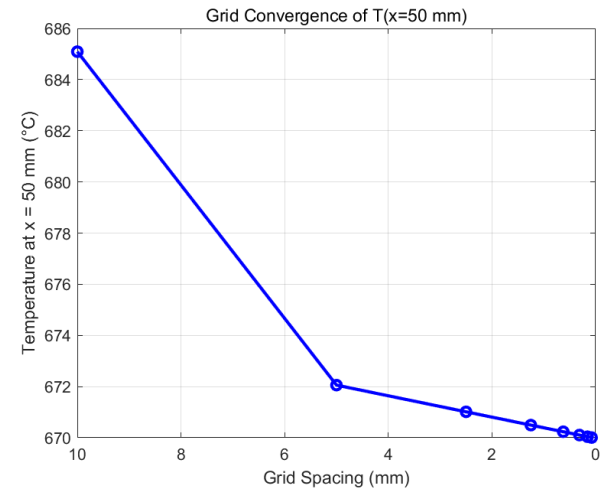
```

89 %% Precomputation for Finite-Volume Discretization
90
91 % Thermal capacity per node (CV), assuming unit cross-sectional area.
92 rcw = rho_arr .* c_arr .* w; % [J/K]
93
94 % Number of faces between nodes
95 N_faces = N_nodes - 1;
96 Ccond_face = zeros(N_faces, 1); % Conduction coefficients at cell faces
97
98 %% Initialization
99 T = T_init * ones(N_nodes, 1); % Set initial temperature at all nodes [K]
100 time_history(1) = 0;
101 T_history(1,:) = T';
102
103 % Initial recording of min and max temperatures in each region (in K)
104 min_conc_T_hist(1) = min(T(concrete_idx));
105 max_conc_T_hist(1) = max(T(concrete_idx));
106 min_steel_T_hist(1) = min(T(steel_idx));
107 max_steel_T_hist(1) = max(T(steel_idx));
108
109 %% Time Marching (Fully Implicit with Semi-Implicit Radiation)
110 step = 0;
111 while t < t_end
112     step = step + 1;
113     t = t + dt;
114
115     % Update conduction coefficients at each face using harmonic mean
116     for f = 1:N_faces
117         kL = k_arr(f);
118         kR = k_arr(f+1);
119         % Distance between cell centers is half the cell width from each side
120         dist = 0.5 * w(f) + 0.5 * w(f+1);
121         k_face = 2 * kL * kR / (kL + kR);
122         Ccond_face(f) = k_face / dist;
123     end
124
125     % Assemble the linear system A * T_new = b_vec
126     A = zeros(N_nodes, N_nodes);
127     b_vec = zeros(N_nodes, 1);
128
129     % Node 1 (x = 0 mm, exposed surface: convection + radiation)
130     i = 1;
131     A(i,i) = rcw(i)/dt + Ccond_face(i) + h;
132     A(i,i+1) = -Ccond_face(i);
133     % Compute radiation term using previous T(i) (semi-implicit)
134     radTerm = epsilon * sigma * ( T(i)^4 - T_inf^4 );
135     b_vec(i) = rcw(i)/dt * T(i) + h*T_inf - radTerm;
136
137     % Internal Nodes (nodes 2 to N_nodes-1)
138     for i = 2:N_nodes-1
139         aW = Ccond_face(i-1);
140         aE = Ccond_face(i);
141         A(i, i-1) = -aW;
142         A(i, i) = rcw(i)/dt + aW + aE;
143         A(i, i+1) = -aE;
144         b_vec(i) = rcw(i)/dt * T(i);
145     end
146
147     % Last Node (x = L_total_mm, adiabatic condition)
148     i = N_nodes;
149     A(i, i-1) = -Ccond_face(i-1);
150     A(i, i) = rcw(i)/dt + Ccond_face(i-1);
151     b_vec(i) = rcw(i)/dt * T(i);
152
153     % Solve the linear system for the new temperature
154     T_new = A \ b_vec;
155     T = T_new;
156
157     % Record time and temperature history
158     time_history(step+1) = t;
159     T_history(step+1,:) = T';
160
161     % Update min and max temperature records for both regions (in K)
162     min_conc_T_hist(step+1) = min(T(concrete_idx));
163     max_conc_T_hist(step+1) = max(T(concrete_idx));
164     min_steel_T_hist(step+1) = min(T(steel_idx));
165     max_steel_T_hist(step+1) = max(T(steel_idx));
166 end
167

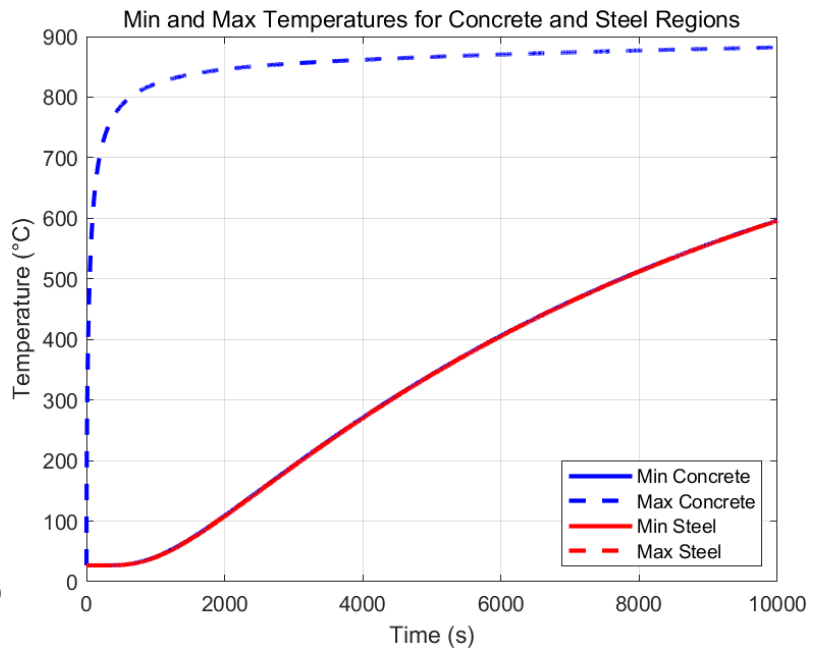
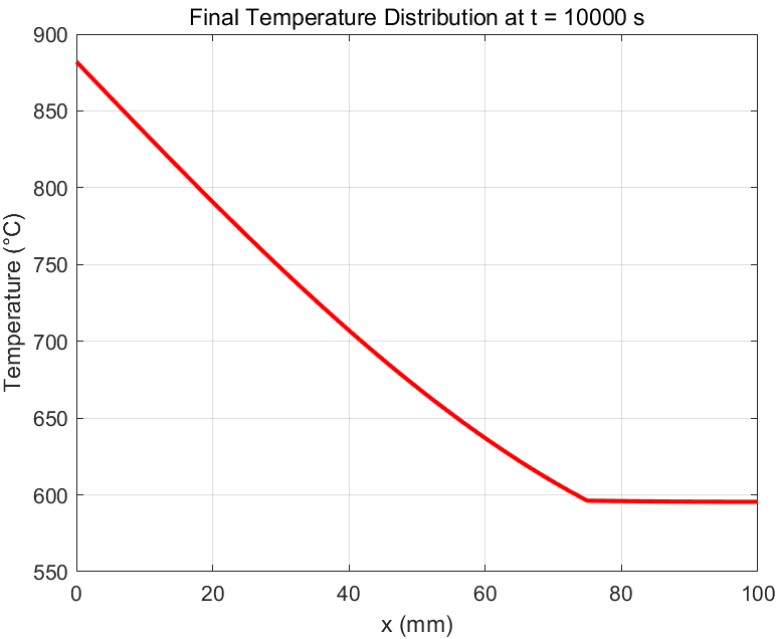
```

thermal
symmetry
condition
applied

* Result and Discussion



```
N_nodes = 11, dx = 10.00 mm, T(x=50 mm) = 685.09 °C
N_nodes = 21, dx = 5.00 mm, T(x=50 mm) = 672.06 °C
N_nodes = 41, dx = 2.50 mm, T(x=50 mm) = 671.01 °C
N_nodes = 81, dx = 1.25 mm, T(x=50 mm) = 670.49 °C
N_nodes = 161, dx = 0.62 mm, T(x=50 mm) = 670.23 °C
N_nodes = 321, dx = 0.31 mm, T(x=50 mm) = 670.11 °C
N_nodes = 641, dx = 0.16 mm, T(x=50 mm) = 670.04 °C
N_nodes = 1281, dx = 0.08 mm, T(x=50 mm) = 670.01 °C
```



```
=== Final Results (t = 10000 s) ===
Exposed Surface (Concrete, x = 0 mm) Temperature : 882.01 °C
Core (Steel, x = 100 mm) Temperature           : 595.50 °C
```

According to the problem conditions, the numerical solution was obtained using the implicit method with $\Delta t = 1$ s and $\Delta x = 7.8125 \times 10^{-5}$ m. The Δx value was chosen as the cell size at which the solution was considered sufficiently converged during the grid test. After 10,000 seconds, the final temperature distribution was calculated to be 882.01 degree celsius at the surface in contact with the fluid. As one moves toward the inner core, the temperature gradually decreases, and in the steel region, the high thermal conductivity results in a nearly constant temperature; in this case, the temperature at the center was calculated to be 595.5 degree celsius. Subsequently, the time-dependent changes in the maximum and minimum temperatures in the concrete and steel regions were examined. The maximum temperature in the concrete region increased the fastest, which is attributed to its direct contact with the fluid. Moreover, since the minimum temperature was calculated at a node close to the interface with the steel, it was nearly identical to the maximum temperature in the steel. In the steel region, due to the high thermal conductivity, the minimum and maximum temperatures varied almost identically over time.